

## Conclusion

## Summary I

This paper has presented `template_pools_rules_tools`, a meta-project exemplar demonstrating how research software projects embedded in a monorepo can integrate three categories of shared resources — data pools (fonds), governance rules, and executable tools — without tight coupling to any specific resource instance. The key contributions are:

## Summary I

1. **A four-module architecture** (`fonds_reader`, `rules_applier`, `tools_invoker`, `integration`) that provides a canonical pattern for resource-aware project code in the template repository. Each module is independently testable and independently deployable.
2. **A typed manifest convention** (`fonds.yaml`, `rules.yaml`, `tools.yaml`) that makes resource capabilities explicit and checkable at pipeline initialisation time, shifting failure detection from runtime to startup — a significant improvement for reproducibility (Wilson et al. 2014).
3. **A token injection pipeline** that links manuscript prose to integration runtime statistics through doubled-brace, uppercase-key placeholders, ensuring that quantitative claims in the manuscript are always generated by the pipeline rather than authored manually. In the current run this covered 3 fonds, 2 rule sets, 4 tools, and 8 bibliography entries.
4. **A three-level resilience design** — resource absence, schema malformation, and script absence — that allows the pipeline to degrade gracefully and report failures informatively rather than crashing, consistent with best practices for robust research software (Taschuk and Wilson 2017).

Several design decisions deserve emphasis as lessons for future exemplar authors:

**Repo-root-relative discovery** is non-negotiable in a forkable monorepo. Any hard-coded absolute path or working-directory assumption will fail when the repository is cloned to a different location. The `pathlib.Path(__file__).resolve().parents[N]` idiom used throughout `src/` ensures that discovery works from any working directory.

**Graceful fallbacks over exceptions** is the correct trade-off for a resource-consumption layer. The alternative — raising `FileNotFoundException` when a fond is missing — would make the integration pipeline fragile to the ordering of parallel agent writes. Returning `None` and logging a warning allows the pipeline to produce a complete, if partial, result dict that downstream consumers can reason about.

**Real-file tests over mocks** is required for a template exemplar. Mocks can pass tests while the real discovery logic is silently broken. Tests that use real file paths exercise the full path-resolution chain and catch regressions that mocks would miss.

**Thin orchestration scripts** keep the scripts directory readable and focus all testable logic in `src/`. A script that does more than parse arguments, call a `src/` function, and write output is accumulating business logic that belongs elsewhere.

## Limitations I

This exemplar makes several deliberate simplifications that a reader adopting the pattern should be aware of:

## Limitations I

- ▶ **Structural, not semantic, validation by default.** `validate_against_rules()` and `discover_tools()` confirm that manifests and rule files are parseable YAML with the expected top-level shape; they do not check that a fond's *data* conforms to its declared schema, or that a tool's entrypoint script actually behaves according to its documented contract. Semantic constraint checking exists only for strong rules, via the separate `strong_rule_evaluator` module (sec. `sec:rules`), and only for the constraint kinds that module implements: `coverage_threshold`, `module_structure`, `section_schema`, `reference_schema` (all four wired to a real strong-rule YAML file and exercised against this project's own live tree), plus `manifest_freshness` (evaluator logic implemented and unit-tested, deliberately not yet wired to a real rule file — see Future Directions).
- ▶ **No concurrency handling.** All readers assume a single-process, single-read invocation. If a parallel agent is actively writing a fond's `data/` file while another process reads it, the reader may observe a partially-written file and report a spurious parse error rather than a clean “not yet available” signal. The architecture tolerates *absence* gracefully; it does not guarantee *atomicity* against concurrent writers.



## Limitations II

- ▶ **Small-scale assumption.** As discussed in sec. `sec:integration's` Performance and Overhead subsection, the design is tuned for a curated, human-reviewed set of resources (single-digit to low-double-digit counts per category), not a resource catalogue at data-lake scale.
- ▶ **English-language, code-review-oriented soft rules.** Soft rules are Markdown prose intended for human reviewers and AI coding agents; they are not evaluated by any automated tool in this exemplar, unlike strong rules. A project that wants soft-rule compliance checked automatically would need to promote the relevant guideline to a strong rule with a corresponding evaluator function.
- ▶ **`strong_rule_evaluator.py` had the thinnest test coverage of any `src/` module, historically.** As measured in an earlier coverage run, it required more negative-control tests (malformed rule files, real violations, type-mismatched context values) than the other source modules, concentrated in its `section_schema`, `reference_schema`, and `module_s` structure evaluators and its context loader. That gap has since been closed with real negative-control fixtures — run `uv run pytest ... --cov-report=term-missing` for the current per-module breakdown, since coverage changes as the suite grows.

**The no-concurrency-handling limitation above is not merely theoretical:** this exemplar is itself designed to be populated by parallel agents authoring `fonds/`, `rules/`, and `tools/` concurrently, which is precisely the scenario where a torn read is most likely. Adopters running this pattern in a genuinely concurrent write environment should add file locking or an atomic-rename write pattern at the resource-authoring layer — this exemplar deliberately does not, to keep the reader-side code minimal.

## Future Directions I

The three-layer architecture described here is deliberately extensible. The most natural extension is a fourth resource category: **models** (`models/<scope>/<name>/`) for pre-trained machine learning models and their inference scripts. The pattern would follow exactly the same manifest-and-reader design, with `models.yaml` declaring type, version, and entrypoints, and a `models_loader` module providing `discover_models()` and `validate_model_files_exist()`. Adding this layer to `template_pools_rules_tools` would require only a new reader module and a new section in the integration orchestrator — the existing architecture places no constraints on the number of resource categories.

A second direction, **cross-fond validation**, is now implemented: `src/integration.py::check_bibliography_overlap()` compares this manuscript's own cite keys against the `template_bibliography` fond's curated set and reports overlap, `project_only`, and `fond_only` counts. It deliberately does *not* assert full containment in either direction — this manuscript legitimately cites software-engineering sources the fond does not curate, and the fond legitimately curates machine-learning sources this manuscript does not cite. Run `uv run python -c "from src.integration import check_bibliography_overlap; import json; print(json.dumps(check_bibliography_overlap(), indent=2))"` from the project root to see the current overlap.

## Future Directions I

A third direction, suggested directly by the Limitations above, was extending `strong_rule_evaluator` with additional rule kinds beyond the original four. The `manifest_freshness` kind — flagging a fond whose version field has not been bumped despite its data/ files changing more recently than its manifest — is now implemented: `src/strong_rule_evaluator.py::_evaluate_manifest_freshness()` is a fifth dispatch entry, unit-tested against synthetic rule dicts (`tests/test_strong_rule_evaluator.py`). It is deliberately **not** yet wired to a real strong-rule YAML file: doing so would mean adding a file under the shared `rules/templates/template_project_rules/strong/` or `rules/templates/template_manuscript_rules/strong/` directory, which this project's own read-only contract against `fonds/`, `rules/`, and `tools/` reserves for an explicit decision

rather than a routine content edit. The evaluator's logic is proven correct against synthetic input; proving it against a real fond's real manifest and data-file timestamps is the natural next step, pending that decision.

The exemplar is ready to fork. A project team wishing to adopt this architecture need only copy the relevant `src/` modules (the three readers plus `integration.py` are the minimum; `strong_rule_evaluator.py` and `figures.py` are optional extensions), update the resource directory paths in each reader, and replace the exemplar `fond/rules/tool` names with their own.